



Response to Request for Information (RFI):

Energy Consumption and Efficiency Solutions

Efovinity is happy to provide a response to the above topic which is one of our core innovation drivers as a company. Our company specializes in developing energy-efficient algorithms and architectures for machine learning and artificial intelligence (AI) systems. Our expertise lies in reducing the energy footprint of computationally intensive processes, such as training large-scale models, while maintaining or improving performance.

We have a proven track record of innovation in this space, including the development of novel architectures for memory-efficient and energy-efficient training of diffusion models on high-dimensional datasets. Our work is particularly relevant to organizations leveraging AI and large language models (LLMs), which are increasingly reliant on energy-intensive infrastructure.

Technology and Solutions

We focus on developing energy-efficient algorithms that reduce the computational and energy demands of AI systems. One of our key innovations is the use of **invertible neural networks** and **graph-based models** to optimize energy usage during training and inference.

Invertible Diffusion Models for Energy Efficiency

We developed a novel architecture for training diffusion models on high-dimensional datasets, such as medical imaging data, using a **single GPU**. This architecture leverages an **invertible UNet** with **invertible attention modules**, which provides two significant contributions: